2023 19th IEEE International Colloquium on Signal Processing & Its Applications (CSPA) | 978-1-6654-7692-8/23/$31.00 ©2023 IEEE | DOI: 10.1109/CSPA57446.2023.10087500

EmHM: A Novel Hybrid Model for the Emotion Recognition based on EEG Signals

1st Ramnivas Sharma
Electrical Engineering Department
Malaviya National Institute of Technology
Jaipur, India
2020ree9534@mnit.ac.in

2nd Hemant Kumar Meena
Electrical Engineering Department
Malaviya National Institute of Technology
Jaipur, India
hmeena.ee@mnit.ac.in

***Abstract*—Emotion is a fundamental aspect of daily life and is crucial for human interactions. This study suggests a unique electroencephalogram (EEG)-based technique for identifying human emotions. For EEG-based emotion analysis, the proposed model is tested on the SEED and DEAP datasets. For the DEAP dataset, we consider valence and arousal emotions for classification purposes, and for the SEED dataset, three emotions, neutral, positive, and negative, have been considered. The differential entropy (DE) is used for the SEED dataset, and for the DEAP dataset, the power spectral density (PSD) is used as a feature. For precise emotion recognition, an EmHM (Emotion Hybrid Model) based on long short-term memory (LSTM) and a convolutional neural network (CNN) are constructed. Furthermore, we applied the CNN, LSTM, and EmHM models, and all three models for emotion recognition are fed with the retrieved information. Various methods improved on already-existing models to accurately classify human emotion. To get better accuracy than the existing techniques, we suggested a model that uses a different approach known as EmHM. By applying all three models such as CNN, LSTM and EmHM, we got the highest accuracy 86.50%, 87.98% and 91.56% respectively on dataset. To improve prediction results, the CNN and the LSTM models are combined to make the EmHM hybrid model.**

***Index Terms*—Emotion recognition, DEAP dataset, CNN, LSTM, SEED datatset, EmHM, power spectral density (PSD)**

I. Introduction

Emotions are feelings that affect one's physical and psychological well-being. There are many emotions, including fear, hatred, happiness, joy, sorrow, fury, pleasure, pain, tenseness, boredom, and so on. Emotions can be expressed through facial expressions, signing, and talking, all of which aid people in working with one another [1], [2]. There are two schools of thought on emotions: one sees emotions as normal states of individuals, whereas the other sees them as physiological interactions [3], [4]. Emotions can be classified using two approaches, such as discrete fundamental emotion classification and the dimension approach [5], [6]. The primary emotions based on discrete fundamentals can be classed as negative and positive emotions. Negative emotions cause us to feel bad. These are the feelings that make us feel let down, unhappy, embarrassed, and powerless. Anger, guilt, anxiety, humiliation, shame, despair, tiredness, panic, frustration, nervousness, regret, irritability, and other negative emotions are examples. Positive emotions provide us with pleasant experiences and thoughts. These are the feelings that energize us, make us happy, make us respect others, and make us feel good about ourselves. Joy, love, faith, enthusiasm, hope, gladness, thrill, pride, and other pleasant feelings are examples. Emotions are classified into three categories using the dimension approach (valence, arousal and dominance) [7], [8]. The degree to which an emotion is negative or positive is referred to as valence. As a result, it is further divided into two categories: fear, tenseness, rage, melancholy, boredom, and so on are examples of low valence or negative valence emotions, while high valence or positive valence emotions are glad, calm, pleased, excited, etc [8], [9].

Arousal [8] is a term that describes the intensity of a state of emotions and reflects the level of enthusiasm, eagerness, or apathy. It's also divided into two groups: Low arousal (Negative) emotions are known as sleepy, bored, calm, melancholy, etc. High arousal (Positive) emotions are defined as annoyed, excited, anxious, energized, joyful, etc. Figure 1 depicts the various types of emotions. The cerebrum, which is divided into left and right hemispheres, is the most essential and most prominent component of the human brain. The frontal, temporal, parietal, and occipital lobes make up each hemisphere. The frontal lobe is in charge of controlling physical movement, personality, focus, and so forth. The temporal lobe regulates emotions such as thirst, hunger, and face recognition, as well as hearing [9], [10]. The parietal lobe controls the sensations of touch, taste, and other senses. Humans can detect colors, form, size, depth, and distance with the help of the occipital lobe [9].

II. Related work

A. Deep learning algorithms approach

There are several ways in the literature for dealing with the EEG emotion categorization problem and different disease detection problems. Some well-known techniques, such as deep neural network (DNN) [7] and CNN [6], [7] algorithms, have been widely used on different characterization problems, including image classification [11], [12], object detection [13], tracking [14], and segmentation. Machine learning (ML) methods have been utilized to handle a variety of complicated challenges in different fields, such as medical, financial, etc. This paper looks at how machine learning is used in medicine,

Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 20,2026 at 04:44:49 UTC from IEEE Xplore. Restrictions apply.

with an emphasis on commonplace technologies and how they impact medical diagnosis. There are major fields such as cancer, medicinal chemistry, the brain, emotion categorization, medical imaging, and wearable sensors in which machine learning models are used for analysis purposes [15]–[17]. In recent scenarios, due to advances in the fields of deep learning, research in the identification of human emotions has accelerated. Different models of a framework for interpreting facial and vocal expressions of emotion exist. This paper presents a computer-aided human emotion interface using deep learning techniques [7] such as CNN, long short-term memory (LSTM) [8] and EmHM (Emotion Hybrid model) based on CNN and LSTM.

B. Contribution of the paper

1) In this paper, a comparative study using different classifiers such as CNN, LSTM, and a EmHM is proposed for emotion classification. The proposed model explores a hybrid model (EmHM) with different deep learning techniques such as CNN and LSTM.
2) The suggested EmHM (CNN-LSTM) is a highly efficient (reduced time and complexity) and automatic recognition model that can be successfully employed for emotion classification on both the DEAP and SEED datasets.
3) A different and comparatively larger DEAP dataset with 32 participants contains EEG and peripheral physiological information and SEED dataset consist 15 participants and 62 channels for each frequency band, the derived DE feature for a sample EEG signal has 310 dimensions are used to train and test the model separately.
4) Pre-processing includes windowing to enable faster model training.
5) This research suggests a model that is substantially less complex since it has fewer hidden layers than other models.

III. Data description

A. DEAP dataset

We use a preprocessed version of DEAP [18]–[20], a widely used and openly accessible EEG dataset, to demonstrate our method. The DEAP dataset contains the EEG and peripheral physiological information of 32 participants or 32 EEG channels. The information gathered from the participants watched 60-second long music videos. Every participant is given 40 trials (music videos) and asked to rate each video on valence, arousal, like/dislike, dominance, and familiarity [19]. As a result, a total of 1280 video ratings were obtained, which was calculated by multiplying the number of videos (40) by the number of volunteers (i.e., 32). EEG data is saved in 32 of the 40 channels available for studies. We focus on valence and arousal classification, and the ratings are decimal integers between 1 and 9, which are further divided into two categories: low and high for the binary classification task. High ratings are greater than or equal to 5, while low ratings are less than 5.

B. SEED dataset

In [21]–[23] SJTU emotion EEG dataset (SEED) was used. The dataset includes EEG data from 15 subjects-7 men and 8 women-who were captured over the course of three distinct sessions, each of which had 15 trials. The EEG signal is captured during each trial as the subject watches Chinese film snippets containing positive, neutral, and negative emotional states. Each movie clip lasts for approximately 4 minutes, and two movies that are meant to evoke the same emotion are not played back to back. Following the viewing of each film clip, the participants filled out a questionnaire to express their feelings about it. Using a 62-channel electrode cap, the EEG data are captured using the worldwide 10-20 system. EEG signal is downsampled from 200 Hz to 0-75 Hz using band-pass filter.

IV. Proposed approach

The CNN model moves through the data in various hidden layers; however, the output in this case is not sent back to the network. CNN algorithms could be better at learning sequential information but strong at extracting temporal aspects. Since LSTM is a component of RNN, it is used for sequential data learning since it not only picks up new information from training but also retains it so it can predict what will come next in the sequence. The method includes feeding the network with the output. As a result, the LSTM processes these features sequentially after learning long-term dependencies (sequence learning). However, to improve emotion classification, a EmHM that combines CNN and LSTM is created and deployed to both datasets.

A. Convolutional Neural Network (CNN)

CNNs have become the standard approach for many practical applications using 2D signals such as image and video frames. CNN may not be a viable solution in many 1D signal applications, especially when training data is sparse or application specific. 1D CNNs were proposed for 1D signal-based applications, and give better performance in different applications such as the classification of biological signals and early diagnosis, monitoring of health conditions, anomaly detection, and electrical motor fault detection. For image processing and type, CNN is the most successful model.

The best part about the CNN design is that no feature extraction is required. Convolution is a mathematical technique used in the CNN model. It uses data convolution and filters instead of normal matrix multiplication to produce a variety of characteristics, which are passed to the next layer [7], [9]. Each layer of the proposed CNN model and the parameters associated with each layer are presented in Table I. Figure 1 represents the basic flowchart of the CNN model for the SEED dataset

B. Long Short term Memory (LSTM)

Long short-term memory (LSTM) is defined as artificial recurrent neural networks (RNNs) of deep learning architecture (DL). RNN used for various applications, such as

Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 20,2026 at 04:44:49 UTC from IEEE Xplore. Restrictions apply.

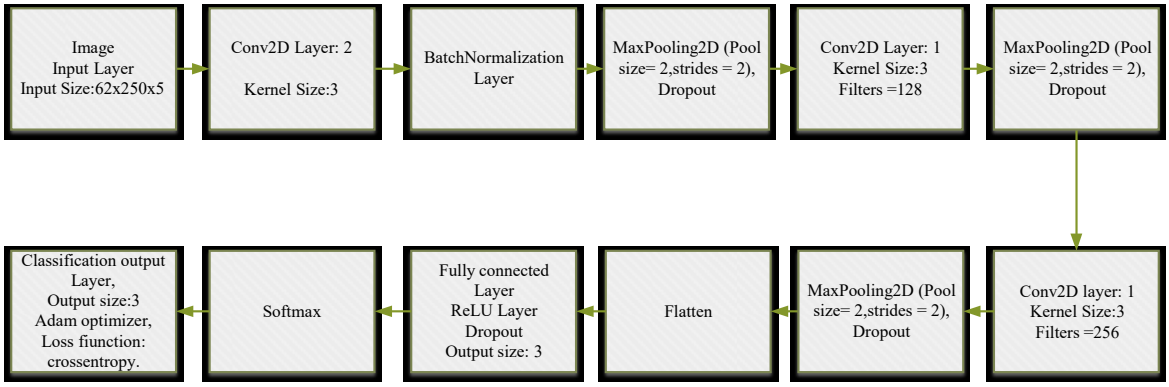

Fig. 1. The basic framework of CNN model

TABLE I
PARAMETERS OF CNN BASED MODEL FOR DEAP DATASET

Name of layers	Pooling	Kernel	Parameters of layer	Dropout	Parameters
Conv1D	Yes (Max pooling)	256x3	ReLU Activation	None	1024
Conv1D	Yes (max pooling	256x3	ReLU Activation	None	196864
Conv1D	Yes (max pooling)	128x3	ReLU activation, Unit size = 64	None	98432
Conv1D	Yes	64x3	ReLU activation, Unit size =64	None	24640
Fully connected	No	None	Activation='tanh', Unit size = 64	0.2	16448
Fully connected	No	None	Activation='tanh', unit size =32	0.2	2080
Fully connected	No	None	Activation= ReLU, Unit size = 16	0.2	528
Fully connected	No	None	Softmax activation, Unit size = 10	None	170

TABLE II
PARAMETERS OF LSTM BASED MODEL FOR DEAP DATASET

S.No	Name of layers	Kernel	Parameters of layers	Dropout	Parameters
1	BLSTM	None	Unit size =256, return sequence = True	0.6	528384
2	LSTM	None	Unit size = 512, return sequence =True	0.6	2099200
3	LSTM	None	Unit size = 128, return sequence =True	0.6	328192
4	LSTM	None	Unit size = 64, return sequence =True	0.4	49408
5	LSTM	None	Unit size = 32, return sequence =True	0.4	12416
6	Fully connected	None	Unit size = 16, ReLU activation	None	528
7	Fully connected	None	Unit size = 10 Softmax activation	None	170

sequential data, text, audio, and video. The LSTM network is used for different tasks such as classifying, processing, and making predictions based on the time series data [7–9]. Early research demonstrated that LSTM could be used to learn a variety of previously impossible-to-learn problems. LSTM has a number of applications based on deep learning, like speech recognition, handwriting recognition, machine translation, and image processing. We also included LSTM in our model and taken different parameter to analyze the model based on it. Each layer of the proposed LSTM model and the parameters associated with each layer is explained in Table II. Figure 2 represents the basic flowchart to create the LSTM model for emotion classification using the datasets.

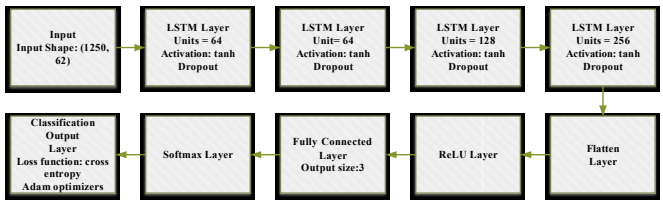

Fig. 2. The basic framework of LSTM model

C. Hybrid model (CNN-LSTM):EmHM

The EmHM (Emotion hybrid model) combines multiple base models such as CNN and LSTM in a single configuration. The basic structure or framework of EmHM is depicted in figure 3. The EmHM boosts performance by gathering additional data that was previously missed. Convolutional (conv) box make up the first three building blocks of the hybrid model consisting the 4 convolutional layer in addition with Maxpool layer and Droupout layer . Max pool layers and Dropout regularization make up the Conv block, which prevents overfitting [24]. The dimension between output of Conv layer and input of LSTM layer are different. Therefor in order to address dimensional mismatch between output layer of Conv Box3 layer and LSTM input layer, the reshape layer is used as shown in figure 3. In LSTM block we used 3 LSTM layer in addition to dropout and activation function. Tanh activation function [25] and batch norm regularisation [26] are both used by the LSTM network. A fully connected network that uses flatten, dense layer and softmax [27] to determine the probabilities of output from the LSTM block.

V. RESULTS DISCUSSION

A. Experimental results for DEAP dataset

Now we will discuss experimental details and results obtained from our implemented model. In our model, we used different types of classifiers such as CNN, LSTM, and EmHM

Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 20,2026 at 04:44:49 UTC from IEEE Xplore. Restrictions apply.

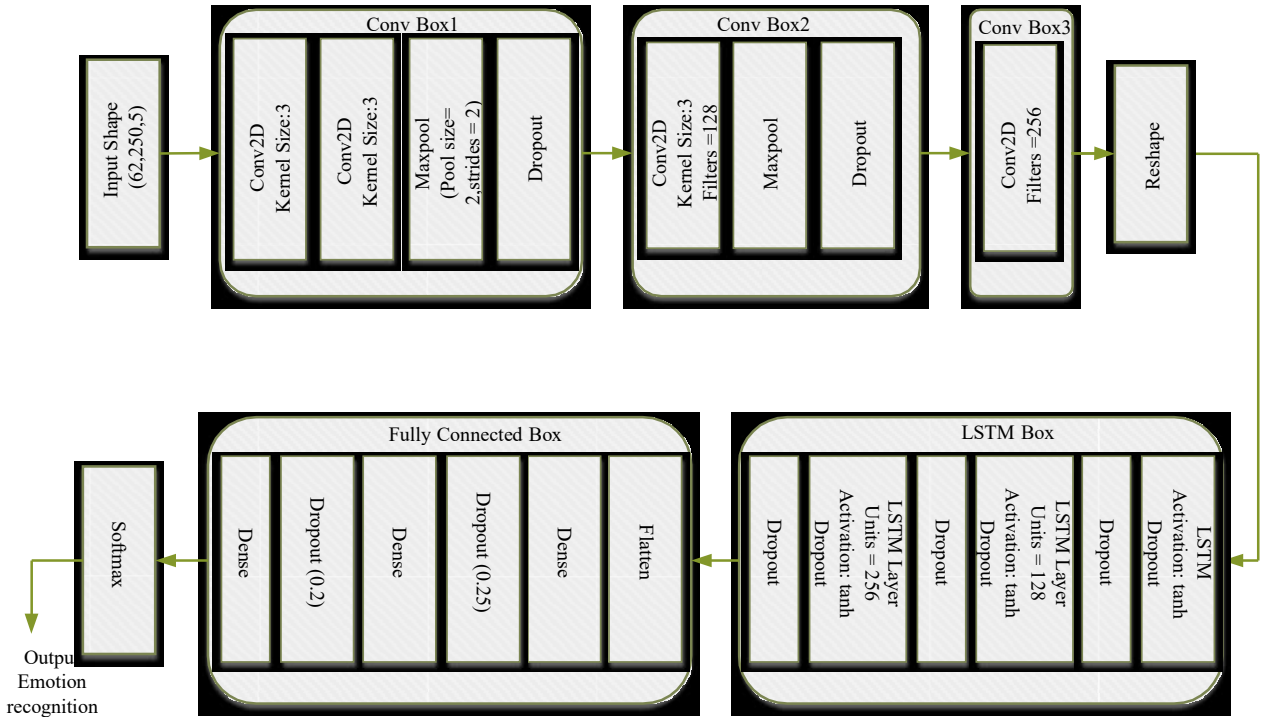

Fig. 3. The basic structure of EmHM (Emotion Hybrid model)

model for emotion classification. But the model accuracy and model loss curve on DEAP dataset with CNN and LSTM architectures are shown in Fig 4 and 5 respectively. The results evaluated with different classifiers are shown in Table V, which indicates the better classification accuracy obtained by EmHM Hybrid model.

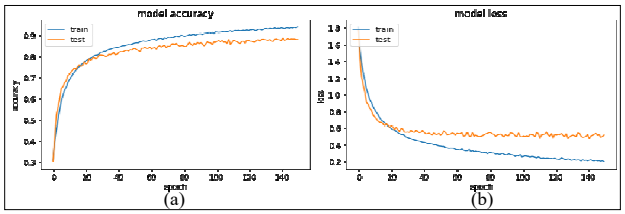

Fig. 4. Accuracy values (a), Loss values (b) curve of CNN model.

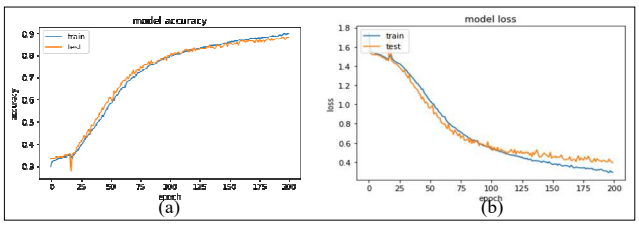

Fig. 5. Accuracy values (a), Loss values (b) curve of LSTM model.

The results of valence and arousal emotion classification is evaluated with accuracy from different techniques. In the CNN and LSTM model, we prefer the 80 – 20 train test split technique. The LSTM model and CNN model have accuracy of 87.68% and 85.23%, respectively.

B. *Experimental results for SEED dataset*

According to the experimental outcomes, the CNN and LSTM models, when used separately, obtain classification accuracy of 81.19% and 84.33% respectively. The accuracy for the EmHM (Emotion hybrid model) is 91.56%. Tables III and IV present the classification report of CNN and the EmHM for SEED data, respectively. Table IV shows that the EmHM model strategy outperforms other models in terms of performance. Figures 6 and 7 show how the accuracy curve and loss function change with time.

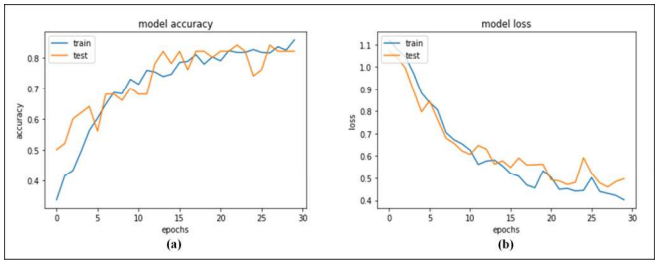

Fig. 6. Accuracy (a), Loss (b) curve of LSTM model of SEED dataset

The performance curve of hybrid model presented in Figure 8 is also known as the accuracy curve. Finally, the results obtained from different implemented models are shown in Table V.

Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 20,2026 at 04:44:49 UTC from IEEE Xplore. Restrictions apply.

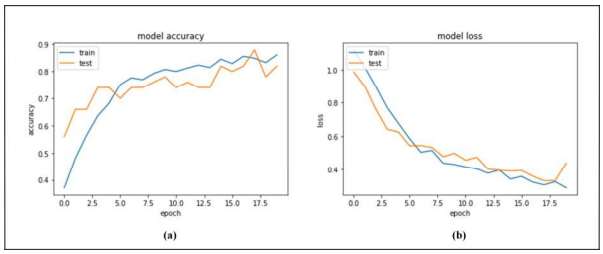

Fig. 7. Accuracy (a), Loss (b) curve of CNN model of SEED dataset

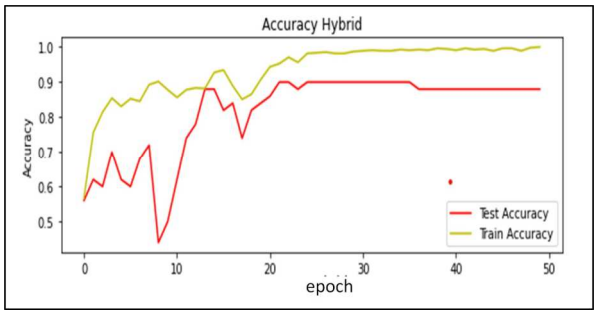

Fig. 8. Accuracy (a), Loss (b) curve of proposed EmHM for SEED dataset

TABLE III
CLASSIFICATION REPORT CNN

	precision	recall	f1-score	support
2	0.72	0.88	0.79	26
0	0.88	0.75	0.81	20
1	0.88	0.79	0.84	29
accuracy			0.81	75
macro avg	0.83	0.81	0.81	75
weighted avg	0.83	0.81	0.81	75

TABLE IV
CLASSIFICATION REPORT OF EMHM MODEL

	precision	recall	f1-score	support
2	0.92	0.88	0.90	26
0	0.89	0.80	0.84	20
1	0.91	0.99	0.95	29
accuracy			0.91	75
macro avg	0.91	0.89	0.90	75
weighted avg	0.91	0.91	0.90	75

VI. CONCLUSION

The EmHM model-based EEG emotion recognition system is proposed in this research. We first extracted the feature DE and PSD of the EEG signals' various frequency bands. These attributes are inputs to the CNN and LSTM models. The CNN and LSTM sub-blocks are combined to create the EmHM (Emotion hybrid model). Based on the CNN and LSTM the EmHM is suggested. The results of the experiments show that the EmHM is better at classifying than the other models used in the proposed method. The proposed EmHM outperforms the compared model for EEG-based emotion identification on the SEED dataset with 91.32% accuracy. When the suggested model is tested on the DEAP dataset with the same features and model settings, it works 89.56% of the time.

TABLE V
PROPOSED WORK RESULTS

Classifiers	Dataset	Accuracy %
CNN	DEAP	85.23 (Valence) 86.50 (Arousal)
	SEED	81.19
LSTM	DEAP	87.68 (Valence) 87.98 (Arousal)
	SEED	84.33
EmHM	SEED	91.56

REFERENCES

[1] M. M. Rahman, A. K. Sarkar, M. A. Hossain, M. S. Hossain, M. R. Islam, M. B. Hossain, J. M. Quinn, and M. A. Moni, "Recognition of human emotions using eeg signals: A review," *Computers in Biology and Medicine*, vol. 136, p. 104696, 2021.

[2] D. Maheshwari, S. K. Ghosh, R. Tripathy, M. Sharma, and U. R. Acharya, "Automated accurate emotion recognition system using rhythm-specific deep convolutional neural network technique with multi-channel eeg signals," *Computers in Biology and Medicine*, vol. 134, p. 104428, 2021.

[3] H. J. Yoon and S. Y. Chung, "Eeg-based emotion estimation using bayesian weighted-log-posterior function and perceptron convergence algorithm," *Computers in biology and medicine*, vol. 43, no. 12, pp. 2230–2237, 2013.

[4] J. Li, H. Hua, Z. Xu, L. Shu, X. Xu, F. Kuang, and S. Wu, "Cross-subject eeg emotion recognition combined with connectivity features and meta-transfer learning," *Computers in Biology and Medicine*, vol. 145, p. 105519, 2022.

[5] M. R. Islam, M. M. Islam, M. M. Rahman, C. Mondal, S. K. Singha, M. Ahmad, A. Awal, M. S. Islam, and M. A. Moni, "Eeg channel correlation based model for emotion recognition," *Computers in Biology and Medicine*, vol. 136, p. 104757, 2021.

[6] W. Lin, C. Li, and S. Sun, "Deep convolutional neural network for emotion recognition using eeg and peripheral physiological signal," in *International conference on image and graphics*. Springer, 2017, pp. 385–394.

[7] S. Tripathi, S. Acharya, R. D. Sharma, S. Mittal, and S. Bhattacharya, "Using deep and convolutional neural networks for accurate emotion classification on deap dataset." in *Twenty-ninth IAAI conference*, 2017.

[8] R. C. Dhingra and S. Ram Avtar Jaswal, "Emotion recognition based on eeg using deap dataset," *European Journal of Molecular & Clinical Medicine*, vol. 8, no. 3, pp. 3509–3517, 2021.

[9] D. Acharya, R. Jain, S. S. Panigrahi, R. Sahni, S. Jain, S. P. Deshmukh, and A. Bhardwaj, "Multi-class emotion classification using eeg signals," in *International Advanced Computing Conference*. Springer, 2020, pp. 474–491.

[10] S. M. Alarcao and M. J. Fonseca, "Emotions recognition using eeg signals: A survey," *IEEE Transactions on Affective Computing*, vol. 10, no. 3, pp. 374–393, 2017.

[11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[12] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Advances in neural information processing systems*, vol. 25, 2012.

[13] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *Advances in neural information processing systems*, vol. 28, 2015.

Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 20,2026 at 04:44:49 UTC from IEEE Xplore. Restrictions apply.

[14] M. Danelljan, A. Robinson, F. Shahbaz Khan, and M. Felsberg, "Beyond correlation filters: Learning continuous convolution operators for visual tracking," in *European conference on computer vision*. Springer, 2016, pp. 472–488.

[15] M. N. Dar, M. U. Akram, R. Yuvaraj, S. G. Khawaja, and M. Murugappan, "Eeg-based emotion charting for parkinson's disease patients using convolutional recurrent neural networks and cross dataset learning," *Computers in Biology and Medicine*, vol. 144, p. 105327, 2022.

[16] M. Shehab, L. Abualigah, Q. Shambour, M. A. Abu-Hashem, M. K. Y. Shambour, A. I. Alsalibi, and A. H. Gandomi, "Machine learning in medical applications: A review of state-of-the-art methods," *Computers in Biology and Medicine*, vol. 145, p. 105458, 2022.

[17] M.-P. Hosseini, A. Hosseini, and K. Ahi, "A review on machine learning for eeg signal processing in bioengineering," *IEEE reviews in biomedical engineering*, vol. 14, pp. 204–218, 2020.

[18] N. Zhuang, Y. Zeng, L. Tong, C. Zhang, H. Zhang, and B. Yan, "Emotion recognition from eeg signals using multidimensional information in emd domain," *BioMed research international*, vol. 2017, 2017.

[19] S. Koelstra, C. Muhl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, and I. Patras, "Deap: A database for emotion analysis; using physiological signals," *IEEE transactions on affective computing*, vol. 3, no. 1, pp. 18–31, 2011.

[20] T. Stajić, J. Jovanović, N. Jovanović, and M. M. Janković, "Emotion recognition based on deap database physiological signals," in *2021 29th Telecommunications Forum (TELFOR)*. IEEE, 2021, pp. 1–4.

[21] R.-N. Duan, J.-Y. Zhu, and B.-L. Lu, "Differential entropy feature for eeg-based emotion classification," in *2013 6th International IEEE/EMBS Conference on Neural Engineering (NER)*. IEEE, 2013, pp. 81–84.

[22] W.-L. Zheng and B.-L. Lu, "Investigating critical frequency bands and channels for eeg-based emotion recognition with deep neural networks," *IEEE Transactions on autonomous mental development*, vol. 7, no. 3, pp. 162–175, 2015.

[23] A. Iyer, S. S. Das, R. Teotia, S. Maheshwari, and R. R. Sharma, "Cnn and lstm based ensemble learning for human emotion recognition using eeg recordings," *Multimedia Tools and Applications*, pp. 1–14, 2022.

[24] S. Park and N. Kwak, "Analysis on the dropout effect in convolutional neural networks," in *Asian conference on computer vision*. Springer, 2016, pp. 189–204.

[25] G. A. Anastassiou, "Multivariate hyperbolic tangent neural network approximation," *Computers & Mathematics with Applications*, vol. 61, no. 4, pp. 809–821, 2011.

[26] T. Van Laarhoven, "L2 regularization versus batch and weight normalization," *arXiv preprint arXiv:1706.05350*, 2017.

[27] W. Liu, Y. Wen, Z. Yu, and M. Yang, "Large-margin softmax loss for convolutional neural networks," *arXiv preprint arXiv:1612.02295*, 2016.

Authorized licensed use limited to: Faculty of Science & Technology IFHE. Downloaded on January 20,2026 at 04:44:49 UTC from IEEE Xplore. Restrictions apply.